OPTICAL-FIBRE CABLE TESTING
Practices And Challenges

Testing evaluates the performance of fibre-optic components, cable plants and systems. For every fibre-optic cable plant, testing for continuity and polarity, end-to-end insertion loss and then troubleshooting any problems is necessary

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(Credit: youtube.com)

An optical-fibre cable, also known as fibre-optic cable, is like an electrical cable but contains one or more optical fibres that are used to carry light (signals). An optical-fibre cable consists of a core and a cladding layer. Signal transmission takes place in this type of cable due to total internal reflection, which happens due to the difference in the refractive index between the fibre material and air or any other surrounding material.

In September 2012, NTT Japan demonstrated a single fibre cable that was able to transfer one petabit per second over 50 kilometres. A single, modern fibre cable contains up to 1000 fibres, with tetrabytes per second of potential bandwidth.

Today, optical communication is preferred over wireless transmission because of its large bandwidth and low propagation attenuation. Also, it costs less.

Advantages of an optical-fibre cable are:

- Speed. A fibre-optic cable network operates at a high speed, in gigabytes.
- Bandwidth. It provides large computing capacity.
- Distance. Signals can be transmitted without needing to be refreshed or strengthened.
- Resistance. It offers greater resistance to electromagnetic noise from radios, motors or other nearby cables.
- Maintenance. It requires extremely low maintenance.
- It is less bulky, lighter and more flexible than conventional cables.